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Road surface drawing control device

Abstract

A road surface drawing control device for a vehicle that comprises a target information acquisition device that acquires information on targets around the vehicle; a communication device that communicates with a keyless entry terminal device operated by a user of the vehicle; a road surface drawing device that draws a road surface around the vehicle; a road surface drawing ECU that controls the operation of the road surface drawing device, and when the road surface drawing ECU determines that the user of the vehicle is within a reference distance from the vehicle based on the target information acquired by the target information acquisition device or that the keyless entry terminal device has been operated based on the communication between the communication device and the keyless entry terminal device, the road surface drawing device is activated to draw the road surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. JP2022-188329 filed on Nov. 25, 2022, the content of which is hereby incorporated by reference in its entirety into this application.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a road surface drawing control device for a vehicle such as an automobile.

2. Description of the Related Art

(3) It is known to search for an own vehicle by operating a remote terminal device such as a remote entry key and flashing hazard lamps in the event that a user has forgotten a parking position of the own vehicle in a large parking lot such as a shopping mall parking lot. For example, an example of a hazard lamp lighting device in which hazard lamps are flashed by operating a remote terminal device is described in Japanese Patent Application Laid-open No. 2006-130937.

(4) However, if hazard lamps are shielded by other vehicles or structures in a parking lot, the own vehicle cannot be discovered even if the hazard lamps are flashed. It is also conceivable to activate a buzzer or the like by operating a remote terminal device, but there is also an effect of reverberation, etc., and it is not easy to identify the vehicle in which the buzzer sound is generated.

SUMMARY

(5) The present disclosure provides a road surface drawing control device that can increase a possibility that a user finds his or her own vehicle by using road surface drawing around the vehicle by a road surface drawing device.

(6) According to the present disclosure, there is provided a road surface drawing control device for a vehicle comprising: a target information acquisition device that acquires information of targets around a vehicle; a communication device that communicates with a remote operation terminal device that is operated by a user of the vehicle; a road surface drawing device that draws a road surface around the vehicle; and an electronic control unit that controls the operation of the road surface drawing device.

(7) The electronic control unit is configured to operate the road surface drawing device to draw the road surface in at least one of a situation where it is determined that the user of the vehicle is within a range of a reference distance from the vehicle and a situation where it is determined that the remote operation terminal device is operated by communication between the communication device and the remote operation terminal device, based on at least one of information of targets acquired by the target information acquisition device and communication between the communication device and the remote operation terminal device.

(8) According to the above-described configuration, when the user of the vehicle is determined to be within the range of the reference distance from the vehicle, or when it is determined that the remote operation terminal device is operated, the road surface drawing device is operated and the road surface drawing is performed. Since the road surface drawing is performed around the vehicle, for example, even if the hazard lamps are shielded by other vehicles or structures in a parking lot, if characters or symbols of the road surface drawing are not shielded, the user can identify own vehicle by finding them. Therefore, it is possible to increase a possibility that the user discovers the own vehicle.

(9) In another aspect of the disclosure, the electronic control unit is configured to operate a road

surface drawing device to draw a road surface and operate hazard lamps of the vehicle, wherein the road surface drawing device includes at least one drawing unit that irradiates light of the road surface drawing, and the drawing unit is provided at a position different from the hazard lamps when viewed from above the vehicle.
(10) In another aspect of the present disclosure, the electronic control unit is configured to operate the road surface drawing device to draw a road surface when it is determined that the user of the vehicle is searching for the vehicle on the basis of at least one of the information of targets acquired by the target information acquisition device and the communication between the communication device and the remote operation terminal device.
(11) Further, in another aspect of the present disclosure, the road surface drawing device includes a plurality of drawing units that irradiate light of the road surface drawing in directions different from each other around the vehicle, and the electronic control unit is configured to determine a direction of the user of the vehicle with respect to the vehicle based on at least one of information of the targets acquired by the target information acquisition device and the communication between the communication device and the remote operation terminal device, and to determine a drawing unit to be operated based on the determined direction.
(12) Further, in another aspect of the present disclosure, the electronic control unit is configured to change recognizability of the road surface drawing according to at least one of a distance between the user of the vehicle and the vehicle, brightness around the vehicle and the number of operations on the remote operation terminal device such that the recognizability of the road surface drawing becomes higher as the distance between the user of the vehicle and the vehicle is larger, the recognizability of the road surface drawing becomes higher as the brightness around the vehicle is higher, and the recognizability of the road surface drawing becomes higher as the number of operations on the remote operation terminal device is larger.
(13) Other objects, other features and attendant advantages of the present disclosure will be readily understood from the description of embodiments of the disclosure described with reference to the following drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic configuration diagram showing an embodiment of a road surface drawing control device according to the present disclosure.
(2) FIG. **2** is a flowchart illustrating a routine of the road surface drawing control in the first embodiment.
(3) FIG. **3** is a flowchart showing a main part of a routine of the road surface drawing control in the second embodiment.
(4) FIG. **4** is a flowchart showing a main part of a routine of the road surface drawing control in the third embodiment.
(5) FIG. **5** is a flowchart showing a main part of a routine of the road surface drawing control in the fourth embodiment.
(6) FIG. **6** is a diagram illustrating a procedure of determining a drawing unit to be activated.
DETAILED DESCRIPTION
(7) It will be described in detail below a road surface drawing control device according to embodiments of the present disclosure with reference to the accompanying drawings.
(8) As shown in FIG. **1**, a road surface drawing controller **100** according to an embodiment of the present disclosure is applied to a vehicle **102** and includes a road surface drawing ECU **10**. The vehicle **102** includes a driving assistance ECU **20** and a meter ECU **30**. ECU means an electronic controller unit comprising a microcomputer as its main part.

(9) The microcomputer of each ECU includes a CPU, a ROM, a RAM, readable and writable non-volatile memories (N/M), interfaces (I/F), and the like. The CPU realizes various functions by executing instructions (programs and routines) stored in the ROM. Furthermore, these ECU are connected to each other data-interchangeable (communicating) via a CAN (Controller Area Network) **104**. Thus, information acquired by sensors (including switches) connected to a particular ECU may be transmitted to other ECUs.
(10) The road surface drawing ECU **10** are connected with a transceiver **12** that wirelessly communicates with a keyless entry terminal device **106**, a front drawing unit **14**F, a right drawing unit **14**R, a left drawing unit **14**L and a rear drawing unit **14**B. The road surface drawing ECU **10** and the drawing units **14**F, **14**R, **14**L and **14**B function as a road surface drawing device **16** that draws a road surface around the vehicle **102**. The transceiver **12** operates even in a situation where an ignition switch (not shown) is off. When the transceiver **12** receives a signal wirelessly transmitted from the keyless entry terminal device **106**, the transceiver turns on the ignition switch so as to activate ECUs such as the road surface drawing ECU **10**. The drawing units **14**F, **14**R, **14**L and **14**B are collectively referred to as a drawing unit **14**.
(11) The keyless entry terminal device **106** includes a locking switch and an unlocking switch and transmits a locking signal and an unlocking signal, respectively when the locking switch and the unlocking switch is operated by a user. When the locking signal and the unlocking signal transmitted from the keyless entry terminal device **106** is received by the transceiver **12**, their signals are transmitted to a vehicle body ECU not shown in FIG. **1** via the road surface drawing ECU **10**, and the vehicle body ECU performs locking and unlocking of doors, not shown.
(12) The road surface drawing ECU **10** controls operation of the drawing unit **14**. In particular, when the transceiver **12** receives the locking signal and the unlocking signal from the keyless entry terminal device **106**, as will be described in detail later, the road surface drawing ECU **10** controls the operation of the drawing units **14**F, **14**R, **14**L and **14**B. Furthermore, when the transceiver **12** receives the locking signal and the unlocking signal from the keyless entry terminal device **106**, the road surface drawing ECU **10** estimates a distance L between the user and the vehicle **102** based on a strength of the signal.
(13) Although not shown in FIG. **1**, each drawing unit, as is well known, includes a light source device for drawing, and by the light source device being controlled by the road surface drawing ECU **10**, irradiates light of the road surface drawing on a road surface around the vehicle **102** to draw images such as characters and symbols. As shown in FIG. **6**, the front drawing unit **14**F is provided at a front end of the vehicle **102** to draw information of changes in a traveling direction of the vehicle, and the like on the road surface in front of the vehicle by irradiating light to the front of the vehicle. The right drawing unit **14**R and the left drawing unit **14**L are provided on right and left door mirrors of the vehicle **102**, respectively, and draw notice information about door opening on the right and left road surfaces of the vehicle by irradiating light to the sides of the vehicle. The rear drawing unit **14**B is provided at a rear end of the vehicle **102**, and draws information such as information of vehicle backward on a road surface behind the vehicle by irradiating light to the rear of the vehicle.
(14) The driving assistance ECU **20** is a central control unit which carries out driving assistance control such as lane departure prevention control and following-up inter-vehicle distance control. In the embodiment, the driving assistance ECU **20** provides the road surface drawing ECU **10** with information required for controlling the operation of the front drawing unit **14**F and the like, as will be more fully described later.
(15) A camera sensor **22** and a radar sensor **24** are connected to the driving assistance ECU **20**. The camera sensor **22** includes four camera sensors that capture front, rear, right and left sides, but is not limited to four. The radar sensor **24** includes five radar sensors that acquire target information of a three-dimensional object present in the front area, the right front area, the left front area, the right rear area, and the left rear area, but is not limited to five. The camera sensor **22** and the radar

sensor **24** function as a target information acquisition device **26** that acquires target information around the own vehicle **102**.

(16) In the embodiment, when the unlocking switch of the keyless entry terminal device **106** is operated, in a situation where the ignition switch is off, the driving assistance ECU **20** and the like are activated, the camera sensor **22** starts capturing around the vehicle **102**. In place of the radar sensor **24**, or in addition to the radar sensor **24**, LiDAR (Light Detection And Ranging) may be used.

(17) Furthermore, the driving assistance ECU **20** is connected with a setting operation unit **28**. The setting operation unit **28** is provided in a position to be operated by a driver. Although not shown in FIG. **1**, the setting operation unit **28** includes various switches necessary for driving assistance control.

(18) In the embodiment, an image of an upper body of the user captured by a vehicle interior camera, which is not shown in FIG. **1**, an image of the upper body of the user and a moving image of the walking user captured by a camera of a smartphone and the like are stored in a storage device of the driving assistance ECU **20**. The CPU of the driving assistance ECU **20** determines whether or not a person around the vehicle **102** captured by the camera sensor **22** is the user based on the images stored in the storage device. The CPU of the driving assistance ECU **20** determines whether or not a person around the vehicle **102** captured by the camera sensor **22** is the user by gait determined based on the moving image stored in the storage device. Furthermore, the CPU of the driving assistance ECU **20**, when it is determined that the person captured by the camera sensor **22** is the user, estimates a distance L between the user and the vehicle **102** based on the images captured by the camera sensor **22**.

(19) The meter ECU **30** is connected with left and right hazard lamps **32**F on the front side of the vehicle, left and right hazard lamps **32**B on the rear side of the vehicle, a buzzer **34** and a light receiving sensor **36** of an auto light. When the unlocking switch of the keyless entry terminal device **106** is operated, the hazard lamps **32**F and **32**B perform answerback by blinking for a preset time. As shown in FIG. **6**, the front left and right hazard lamps **32**F are provided on left and right corners of the front end of the vehicle **102**, the rear left and right hazard lamps **32**B are provided on left and right corners of the rear end of the vehicle **102**. Therefore, the drawing units **14**F, **14**R, **14**L and **14**B are provided at positions that differ from the positions of the hazard lamps **32**F and **32**B when viewed from above. However, the drawing unit **14** may include a drawing unit provided at the same position as the hazard ramp **32**F or **32**B.

(20) The buzzer **34** sounds when calling attention to an occupant or occupants of the vehicle, and sounds as needed when road surface drawing is performed by the front side drawing unit **14**F or the like. The sounding sound of the buzzer **34** is also perceivable by persons outside the vehicle **102**. The light receiving sensor **36** is disposed in the vicinity of a windshield (not shown) in a vehicle interior of the vehicle **102**, detects an amount of light Q as an indicator of brightness outside the vehicle.

First Embodiment

(21) In the first embodiment, the ROM of the road surface drawing ECU **10** stores a program of the road surface drawing control corresponding to the flowchart shown in FIG. **2**. The CPU of the road surface drawing ECU **10** is activated when the transceiver **12** receives a signal that is wirelessly transmitted from the keyless entry terminal device **106**, and executes the road surface drawing control according to the program. Incidentally, when it is determined that the vehicle **102** is parked in a place such as a large parking lot, for example, by a navigation device (not shown), the ECUs such as the road surface drawing ECU **10** are energized at each predetermined time, and the road surface drawing control is executed for a preset duration. The same goes in other embodiments described later.

(22) <Road Surface Drawing Control Routine in the First Embodiment>

(23) Next, a routine of the road surface drawing control in the first embodiment will be described

with reference to the flowchart shown in FIG. **2**. In the following description, the road surface drawing control is simply referred to as “the present control”. Incidentally, in FIG. **2** and FIGS. **3** to **5** described later, the road surface drawing is denoted as RSD, and the answerback is denoted as AB.

(24) First, in step S**10**, the CPU determines whether or not the road surface drawing is being performed. When an affirmative determination is made, the present control proceeds to step S**110**, and when a negative determination is made, the present control proceeds to step S**20**.

(25) In step S**20**, the CPU determines whether or not an answerback is being performed, that is, whether or not the hazard lamps **32**F and **32**B are blinking. When a negative determination is made, the present control proceeds to step S**50**, and when an affirmative determination is made, the present control proceeds to step S**30**.

(26) In step S**30**, the CPU determines whether or not an end condition of the answerback is satisfied. When a negative determination is made, the present control temporarily ends, and when an affirmative determination is made, the present control proceeds to step S**40**. For example, when any one of the following conditions X1 to X3 is determined to be satisfied, it may be determined that the end condition of the answerback is satisfied. (X1) Time more than a reference elapsed time (a positive constant) has elapsed since the answerback was started in step S**70**. (X2) The user has moved within a reference end distance (a positive constant) from the vehicle **102**. (X3) One of doors of the vehicle **102** is opened.

(27) In step S**40**, the CPU terminates the answerback to a user's unlock request. Even when the answerback is terminated, the doors of the vehicle **102** are maintained in an unlocked state.

(28) In step S**50**, the CPU determines whether or not the user has approached the vehicle within a reference approach distance (a positive constant greater than the reference end distance) based on a distance L between the user and the vehicle **102** estimated by the CPU and the CPU of the driving assistance ECU **20**. When an affirmative determination is made, the present control proceeds to step S**80** and when a negative determination is made, the present control proceeds to step S**60**.

(29) In step S**60**, the CPU determines whether or not the transceiver **12** has received the unlocking signal transmitted from the keyless entry terminal device **106**. When a negative determination is made, the present control temporarily ends, and when an affirmative determination is made, the present control proceeds to step S**70**. Alternatively, when an affirmative determination is made, the present control may proceed to step S**100**.

(30) In step S**70**, the CPU issues an answerback command to the meter ECU **30**. Upon receipt of an answerback command, the meter ECU **30** initiates an answerback to the user's unlock request by initiating flashing of front and rear hazard lamps **32**F and **32**B.

(31) In step S**80**, the CPU, as in step S**60**, determines whether or not the transceiver **12** has received the unlocking signal transmitted from the keyless entry terminal device **106**. When an affirmative determination is made, the present control proceeds to step S**100**, and when a negative determination is made, the present control proceeds to step S**90**. Incidentally, when a negative determination is made, it may be determined whether or not the user is searching for the vehicle **102** in the same manner as in step S**170** described below, and when an affirmative determination is made, the present control may proceeds to step S**90**, and when a negative determination is made, the present control may once end.

(32) In step S**90**, the CPU initiates the road surface drawing by the drawing unit to a surrounding area of the vehicle **102** by initiating outputting of control signal to the drawing units **14**F, **14**R, **14**L and **14**B.

(33) In step S**100**, the CPU initiates the road surface drawing to the surrounding area of the vehicle **102** by the drawing units **14**F, **14**R, **14**L and **14**B as in step S**90**, and also initiates the answerback to the user's unlock request as in step S**70**. Incidentally, in steps S**70** and S**100**, the doors of the vehicle **102** are unlocked. In addition, when there is an object such as a wall, another vehicle, or a traffic participant, which is an obstacle to the road surface drawing, the road surface drawing on the

side of the object or the traffic participant may not be performed.
(34) In step S**110**, the CPU determines whether or not the answerback is being performed. When a negative determination is made, the present control proceeds to step S**160**, and when an affirmative determination is made, the present control proceeds to step S**120**.
(35) In step S**120**, the CPU determines whether or not end conditions of the road surface drawing and the answerback are satisfied. When a negative determination is made, the present control proceeds to step S**140**, and when an affirmative determination is made, the present control proceeds to step S**130**. For example, when any one of the following conditions Y1 to Y3 is determined to be satisfied, it may be determined that the end conditions of the road surface drawing and the answerback are satisfied. (Y1) Time more than a standard elapsed time has elapsed since the road surface drawing and the answerback are initiated in step S**100**. The reference elapsed time for the road surface drawing may be longer than the reference elapsed time for the answerback. (Y2) The user has moved to within the reference end distance from the vehicle **102**. (Y3) One of the doors of the vehicle **102** is opened.
(36) In step S**130**, the CPU terminates the road surface drawing around the vehicle **102** by the drawing units **14**F, **14**R, **14**L and **14**B and terminates the answerback to the user's unlock request. Even when the answerback is terminated, the doors of the vehicle **102** are maintained in the unlocked state.
(37) In step S**140**, the CPU determines whether or not the user is searching for the vehicle **102**. When an affirmative determination is made, the present control once ends, and when a negative determination is made, the present control proceeds to step S**150**. If, for example, it is determined that a person who is determined to be the user by the CPU of the driving assistance ECU **20** is changing a direction of a line of sight without moving or is moving without directly approaching the vehicle **102**, it may be determined that the user is searching for the vehicle **102**.
(38) In step S**150**, the CPU terminates the road surface drawing around the vehicle **102** by the drawing units **14**F, **14**R, **14**L and **14**B, but continues an answerback to the user's unlock request.
(39) In step S**160**, the CPU determines whether or not the end condition of the road surface drawing is satisfied. When an affirmative determination is made, the present control proceeds to step S**180**, and when a negative determination is made, the present control proceeds to step S**170**. For example, when any one of the following conditions Z1 to Z3 is determined to be satisfied, it may be determined that the end condition of the road surface drawing is satisfied. (Z1) Time more than the standard elapsed time has elapsed since the road surface drawing is initiated in step S**90**. (Z2) The user has moved to within the reference end distance from the vehicle **102**. (Z3) One of the doors of the vehicle **102** is opened.
(40) In step S**170**, the CPU determines whether or not the user is searching for the vehicle **102**, as in step S**140**. When an affirmative determination is made, the present control once ends, and when a negative determination is made, the present control proceeds to step S**180**.
(41) In step S**180**, the CPU terminates the road surface drawing around the vehicle **102** by the drawing units **14**F, **14**R, **14**L and **14**B.
Second Embodiment
(42) FIG. **3** is a flowchart showing a main part of a routine of the road surface drawing control in the second embodiment. In the second embodiment, steps other than steps S**210** to S**250** are performed in the same manner as in the first embodiment. As illustrated in FIG. **3**, however, when an affirmative determination is made in the step S**50**, the present control proceeds to step S**210**.
(43) In step S**210**, the CPU determines whether or not the distance L between the user and the vehicle **102** is greater than a first reference distance L**1** (a positive constant). When a negative determination is made, the present control proceeds to step S**230** and when an affirmative determination is made, a correction coefficient Ka for correcting illuminance of the light source device of the drawing unit **14** is set to 1.2 in the step S**220**.
(44) In step S**230**, the CPU determines whether or not the distance L between the user and the

vehicle **102** is greater than a second reference distance L**2** (a positive constant that is less than the first reference distance L**1**). When a negative determination is made, the correction coefficient Ka is set to 0.8 in step S**240**, and when an affirmative determination is made, the correction coefficient Ka is set to 1 in step S**250**.

(45) In steps S**90** and S**100**, the illuminance of the light source device of the drawing unit **14** when the road surface drawing to the surrounding area of the vehicle **102** is started is corrected to the correction coefficient Ka times. Notably, so long as the correction coefficient Ka set in step S**220** is larger than the correction coefficient Ka set in step S**250**, and the correction coefficient Ka set in step S**240** is smaller than the correction coefficient Ka set in step S**250**, the correction coefficient Ka may not be the above-described value.

Third Embodiment

(46) FIG. **4** is a flowchart showing a main part of a routine of the road surface drawing control in the third embodiment. In the third embodiment, steps other than steps S**310** to S**350** are performed in the same manner as in the first embodiment. As illustrated in FIG. **4**, when an affirmative determination is made in step S**50**, the present control proceeds to step S**310**.

(47) In step S**310**, the CPU determines whether or not the light amount Q detected by the light receiving sensor **36** is greater than a first reference distance Q**1** (a positive constant). When a negative determination is made, the present control proceeds to step S**330**, and when an affirmative determination is made, a correction coefficient Kb for correcting the illuminance of the light source device of the drawing unit **14** is set to 1.2 in step S**320**.

(48) In step S**330**, the CPU determines whether or not the light amount Q detected by the light receiving sensor **36** is greater than a second reference distance Q**2** (a positive constant smaller than the first reference distance Q**1**). When a negative determination is made, the correction coefficient Kb is set to 0.8 in step S**340**, and when an affirmative determination is made, the correction coefficient Kb is set to 1 in step S**350**.

(49) In steps S**90** and S**100**, the illuminance of the light source device of the drawing unit **14** when the road surface drawing to the surrounding area of the vehicle **102** is initiated is corrected to the correction coefficient Kb times. Notably, so long as the correction coefficient Kb set in step S**320** is larger than the correction coefficient Kb set in step S**350**, and the correction coefficient Kb set in step S**340** is smaller than the correction coefficient Kb set in step S**350**, the correction coefficient Kb may not be the above-described value.

Fourth Embodiment

(50) FIG. **5** is a flowchart showing a main part of a routine of the road surface drawing control in the fourth embodiment. In the fourth embodiment, steps other than steps S**410** to S**490** are performed in the same manner as in the first embodiment. As illustrated in FIG. **5**, when an affirmative determination is made in step S**50**, the present control proceeds to step S**410**.

(51) In step S**410**, the CPU determines whether or not the user is in front of the vehicle **102**, for example, by determining whether or not the user is in a region Af illustrated in FIG. **6**. When a negative determination is made, the present control proceeds to step S**430** and when an affirmative determination is made, the drawing unit **14** to be activated is determined to the front drawing unit **14**F in step S**420**.

(52) In step S**430**, the CPU determines whether or not the user is to the right of the vehicle **102**, for example, by determining whether or not the user is in a region Ar shown in FIG. **6**. When a negative determination is made, the present control proceeds to step S**450**, and when an affirmative determination is made, the drawing unit **14** to be activated is determined to the right drawing unit **14**R in the step S**440**. In FIG. **6**, reference numeral **15** denotes a road surface drawing by the side drawing unit **14**R. The road surface drawing **15** is a drawing that announces that a door of the driver's seat opens, and moves in a direction away from the vehicle intermittently, as a symbol of the door is indicated by an arrow.

(53) In step S**450**, the CPU determines whether or not the user is on the left side of the vehicle **102**,

for example, by determining whether or not the user is in a region AI shown in FIG. **6**. When a negative determination is made, the present control proceeds to step S**470**, and when an affirmative determination is made, the drawing unit **14** to be activated is determined to the left drawing unit **14**L in step S**460**.

(54) In step S**470**, the CPU determines whether or not the user is to the back of the vehicle **102**, for example, by determining whether or not the user is in a region Ab illustrated in FIG. **6**. When an affirmative determination is made, the drawing unit **14** to be activated is determined to the rear drawing unit **14**B in step S**480**, and when a negative determination is made, the drawing unit **14** to be activated is determined to all of the drawing units **14**F, **14**R, **14**L and **14**B in step S**490**.

(55) As can be seen from the above explanation, according to the embodiments, in a situation in which neither road surface drawing nor answerback is performed (S**10**, S**20**), when it is determined that the user has approached within the range of the reference approach distance from the vehicle **102** (S**50**), the road surface drawing is started (S**90**, S**100**). Then, the road surface drawing is continued until it is determined that the end condition of the road surface drawing is satisfied (S**160**). Since the road surface drawing is performed around the vehicle, for example, even if the hazard lamps are shielded by other vehicles or structures in a parking lot, if the road surface drawing is not shielded, the user can identify the own vehicle **102** by finding the road surface drawing. Therefore, as compared to where the road surface drawing is not performed, it is possible to increase the possibility that the user finds the own vehicle.

(56) Further, according to the embodiments, when it is determined that the user has approached within the range of the reference approach distance from the vehicle **102** (S**50**) and received the unlocking signal transmitted from the keyless entry terminal device **106** (S**80**), the road surface drawing and answerback are initiated (S**100**). Then, the road surface drawing and the answerback are continued until it is determined that the end condition of the road surface drawing is satisfied (S**120**). Therefore, as compared to where only the road surface drawing is performed, it is possible to increase the possibility that the user finds the own vehicle **102**.

(57) In particular, according to the second embodiment, when the distance L between the user and the vehicle **102** is large, the illuminance of the light source device of the drawing unit **14** is increased, and when the distance L is small, the illuminance of the light source device is lowered. That is, the recognizability of the road surface drawing is changed according to the distance L. Therefore, as compared to where the illuminance of the light source device is constant regardless of the distance L, when the user is in a position far from the vehicle, it is easy for the user to notice the road surface drawing, and when the user is in a position close to the vehicle, an energy consumption by the road surface drawing can be reduced.

(58) Further, according to the third embodiment, when the brightness outside the vehicle is high, the illuminance of the light source device of the drawing unit **14** is increased, and when the brightness outside the vehicle is low, the illuminance of the light source device is lowered. That is, the recognizability of the road surface drawing is changed according to the brightness outside the vehicle. Therefore, as compared to where the illuminance of the light source device is constant regardless of the brightness outside the vehicle, when the brightness outside the vehicle is high, the user can easily notice the road surface drawing, and when the brightness outside the vehicle is low, the energy consumption by the road surface drawing can be reduced.

(59) Further, according to a fourth embodiment, a direction of the user with respect to the vehicle **102** is determined, and a drawing unit to be activated is determined based on the determined direction. That is, the drawing unit **14** on the user's side with respect to the vehicle **102** is activated, and the other drawing units are not activated. Therefore, it is possible to reduce the energy consumed by the road surface drawing as compared to where all the drawing units **14**F, **14**R, **14**L and **14**B are always activated while ensuring a situation in which the user is easily aware of the road surface drawing.

(60) Incidentally, when the drawing unit **14** on the side where an object such as a wall, another

vehicle, or a traffic participant exists is not activated, which becomes an obstacle to the road surface drawing, it is possible to prevent the wasteful road surface drawing from being performed and prevent the road surface drawing from adversely affecting the traffic participant.
(61) While the present disclosure has been described in detail with reference to specific embodiments, it will be apparent to those skilled in the art that the present disclosure is not limited to the embodiments described above, and various other embodiments are possible within the scope of the present disclosure.
(62) For example, the reference elapsed time in step S**120** and S**160** may be constant, but it may be variably set so that the longer the number of times the keyless entry terminal device **106** is operated prior to an affirmative determination is made in step S**120** and S**160**. Further, the more the number of times the keyless entry terminal device **106** is operated, the illuminance of the light source device may be increased. The longer the reference elapsed time and the higher the illuminance of the light source device, the higher the recognizability of the road surface drawing. Furthermore, the number of actuated devices such as a buzzer **34**, headlights (not shown) may be increased.
(63) Further, in the above embodiments, although the remote operation terminal device is a keyless entry terminal device **106**, it may be a smartphone capable of communicating with the transceiver **12** by Bluetooth (registered trademark) or the Internet.
(64) Further, although the second to fourth embodiments described above are mutually independent, the embodiments may be implemented in any combination. In particular, when the second and third embodiments are combined and implemented, in steps S**90** and S**100**, the illuminance of the light source device of the drawing unit **14** is corrected to a Ka×Kb times.

Claims

1. A road surface drawing control device for a vehicle comprising: a target information acquisition device that acquires information of targets around a vehicle; a communication device that communicates with a remote operation terminal device that is operated by a user of the vehicle; a road surface drawing device that draws a road surface around the vehicle; and an electronic control unit that controls operation of the road surface drawing device, the electronic control unit is configured to operate the road surface drawing device to draw the road surface in at least one of a situation where the electronic control unit determines that the user of the vehicle is within a range of a reference distance from the vehicle and a situation where the electronic control unit determines that the remote operation terminal device is operated by communication between the communication device and the remote operation terminal device, based on at least one of the information of targets acquired by the target information acquisition device and communication between the communication device and the remote operation terminal device, wherein the electronic control unit is configured to change recognizability of the road surface drawing according to at least one of a distance between the user of the vehicle and the vehicle, brightness around the vehicle and the number of operations on the remote operation terminal device such that the recognizability of the road surface drawing becomes higher as the distance between the user of the vehicle and the vehicle is larger, the recognizability of the road surface drawing becomes higher as the brightness around the vehicle is higher, and the recognizability of the road surface drawing becomes higher as the number of operations on the remote operation terminal device is larger.
2. The road surface drawing control device for a vehicle according to claim 1, wherein the electronic control unit is configured to operate a road surface drawing device to draw a road surface and operate hazard lamps of the vehicle, wherein the road surface drawing device includes at least one drawing unit that irradiates light of the road surface drawing, and the drawing unit is provided at a position different from the hazard lamps when viewed from above the vehicle.
3. The road surface drawing control device for a vehicle according to claim 1, wherein the electronic control unit is configured to operate the road surface drawing device to draw a road

surface when the electronic control unit determines that the user of the vehicle is searching for the vehicle on the basis of at least one of the information of targets acquired by the target information acquisition device and the communication between the communication device and the remote operation terminal device.

4. The road surface drawing control device for a vehicle according to claim 1, wherein the road surface drawing device includes a plurality of drawing units that irradiate light of the road surface drawing in directions different from each other around the vehicle, and the electronic control unit is configured to determine a direction of the user of the vehicle with respect to the vehicle based on at least one of information of the targets acquired by the target information acquisition device and the communication between the communication device and the remote operation terminal device, and to determine a drawing unit to be operated based on the determined direction.